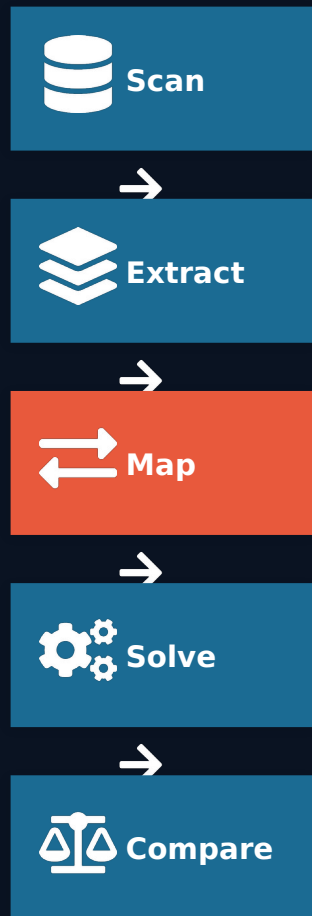


REGULATORY INTELLIGENCE

as External Context Variables for
Supply Chain Optimization

Acme (France) Consultants · Strategic Watch

France · French Africa · MENA



The problem

Supply chain LP/MILP models rely on cost coefficients, capacity bounds, and lead times — but these parameters go stale when regulations change.



Stale parameters

Cost and lead-time assumptions lag behind regulatory shifts by months



Multi-jurisdiction complexity

France, Francophone Africa, MENA — each with distinct, fast-changing regimes



Manual monitoring

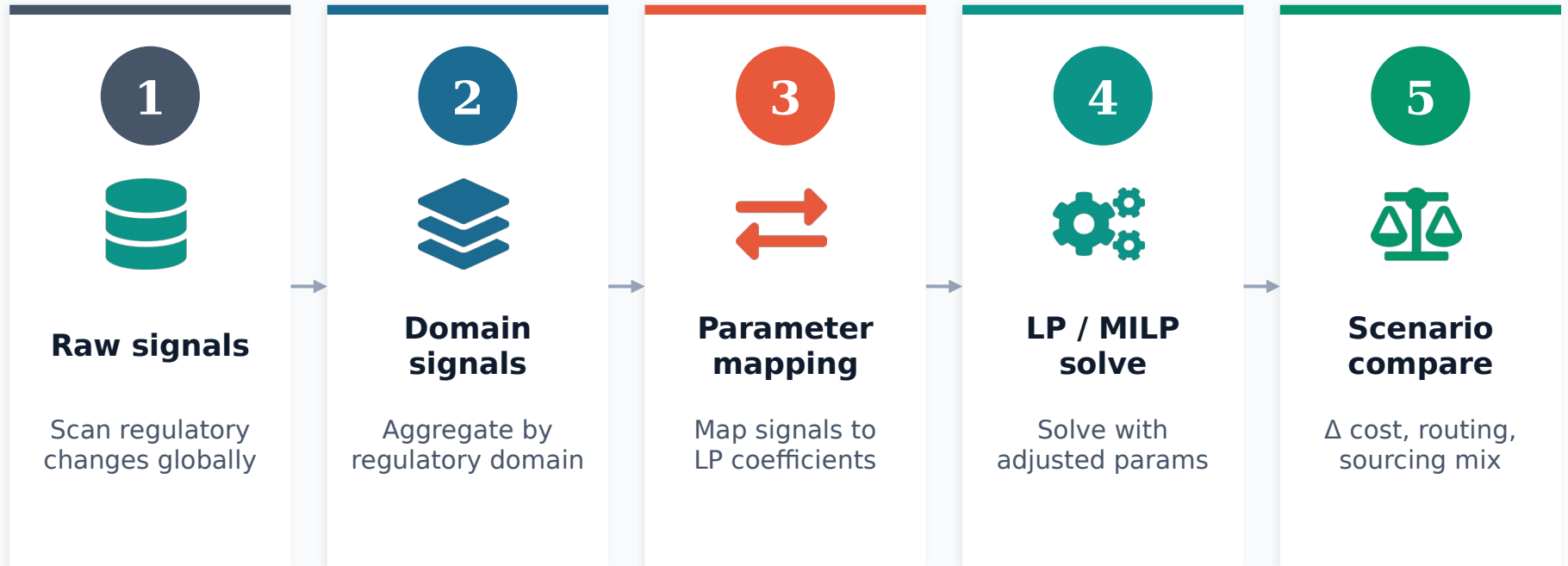
Analysts track 600+ regulators. High-impact changes get missed or arrive late

106,080

regulatory artifacts scanned · 600+ regulators · 12-month lookback

Five-stage pipeline

From raw regulatory updates to optimized supply chain decisions



1 Raw signals — scan annotations

Each regulator produces regulatory updates (press releases, advisories, rule changes). We scan batch files to build a profile per regulator.

Per-update data captured

pub_date — publication timestamp
score_impact (0-10) — severity of change
score_urgency (0-10) — time sensitivity
score_relevance — client relevance
update_type — press release, advisory, rule
tags[] — topic labels (AML, Trade, Cyber...)
impacted_functions[] — Compliance, Ops...

Per-regulator scoring

Frequency (30%)

Log-normalized update count
from raw data — not LLM

Geo fit (30%)

LLM-scored: authority over
France / French Africa / MENA

Client fit (40%)

LLM-scored: relevance to client's
industry and business functions

2

Domain signal extraction

Aggregate updates by regulatory domain into monthly composite signals:
volume × average impact, weighted by supply chain relevance.

Trade & customs

DG Trade, customs bodies,
export controls

$\text{vol} \times \text{avg_impact} / \text{week}$

Financial & AML

AMF, Banque de France,
FinCEN, AML bodies

$\text{vol} \times \text{avg_impact} / \text{week}$

Food & health

DG SANTE, HAS, FDA,
food safety agencies

$\text{vol} \times \text{avg_impact} / \text{week}$

Cyber & data

ANSSI, CNIL, data
protection authorities

$\text{vol} \times \text{avg_impact} / \text{week}$

Output

One numeric timeseries per domain per week. Each point = composite regulatory pressure.
Sliceable by geography (geo_fit), function (Compliance, Operations), or regulator type.

3

Parameter mapping — the bridge

Threshold-triggered rules convert domain signals into LP parameter adjustments.
Calibrated against historical disruption events.

Trade signal ↑

`cost_import[corridor] × 1.15`
`lead_time[corridor] × 1.2`

Financial signal ↑

`txn_cost[route] × 1.08`
`min_compliance_staff += 1`

Food/health signal ↑

`capacity[port] × 0.85`
`inspection_days[origin] += 3`

Cyber signal ↑

`IT_cost[node] × 1.10`
`digital_corridor_eligible = false`

Thresholds calibrated from historical disruption events — not arbitrary.

4

5

Solve & compare scenarios

$$\min \sum (\text{cost}_{ij} \times x_{ij}) + \sum (\text{compliance}_k \times z_k)$$

s.t. capacity constraints
lead time constraints
corridor eligibility

← *adjusted by domain signals*



Baseline

Current regulatory
parameters



Stress

High regulatory
signal scenario



Relaxed

Harmonization
trend scenario

Decision output

$\Delta \text{cost} \cdot \Delta \text{routing} \cdot \Delta \text{sourcing mix}$ across scenarios → actionable supply chain planning

Regulatory intelligence meets supply chain math.

The data doesn't enter the LP as a variable.
It conditions the LP's assumptions — so the model reflects
the world as it is now, not as it was when parameters were last
set.